

# Perfect-correction Markov-channel and continuous Lindblad dynamics at a class-A domain wall

Internal campaign summary<sup>1</sup>

<sup>1</sup>*Class-A fermionic adaptive-circuit project*

(Dated: Production-results note; completed August 22, 2026)

We report a matched comparison between the deterministic trajectory-averaged perfect-correction Markov channel and an infinitesimal continuous Lindblad evolution built from identical overcomplete Wannier modes. Both domain-wall support truncation settings and two finite spatial shells are included. The physical channel retains its number-measurement dephasing; the continuous evolution is evaluated with and without the corresponding number-dephasing term. Because a random ordered sweep need not be translation invariant along the wall, its momentum-resolved estimator is defined by a late-time temporal average followed by an exact translation-group twirl. All three dynamics exhibit a wall-associated natural-occupation crossing while their bulk occupations remain far from one half. The crossing disappears across the parent-model reference near  $\alpha_{\text{run},\text{in}} = 2$ , but no relaxation-gap closure is observed. The continuous density-kick response has opposite signed velocities on the two walls; the ordered channel instead has poor linear fits and a common drift sign. Thus the campaign demonstrates mixed-state occupation spectral flow, not a gapless Lindbladian, trajectory Lyapunov mode, or schedule-independent chiral transport.

## I. CORRECTION AND SCOPE

The computed objects must be distinguished at the outset. The first is the ordered one-cycle correlation-matrix map produced by `classA.U1FGTN.run_markov_channel`. It is the deterministic average of the projective measurement and perfect-feedback protocol, not a sampled trajectory. The second is the continuous equation obtained from infinitesimal local channel increments. The third is the older auxiliary Gaussian bath calculation. At ancillary filling  $\bar{n}_a = 1/2$ , its dephasing-free generator is exactly one half of the perfect-correction dephasing-free continuous generator. Its stationary correlation matrix is therefore reusable, whereas its relaxation clock, response fits, and finite Strang map are not.

No result below is called a trajectory Lyapunov spectrum, a Hamiltonian band, or a Liouvillian many-body spectrum. Natural occupations are eigenvalues of a two-point correlation matrix. A midpoint crossing can diagnose a state-level purity gap without implying a vanishing relaxation rate.

## II. COMMON OW GEOMETRY

We follow the notation of Bhuiyan, Pan, and Jian [1]. Many-body operators carry hats,  $\mu = 1, 2$  labels physical orbitals,  $\nu = A, B$  labels trial-orbital families, and  $\pm$  labels upper and lower parent bands. The parent single-particle matrix in the archived run convention is

$$h_{\alpha_{\text{run}}}(\mathbf{k}) = \hat{\mathbf{n}}_{\alpha_{\text{run}}}(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (1)$$

$$\mathbf{n}_{\alpha_{\text{run}}}(\mathbf{k}) = (\sin k_x, \sin k_y, \alpha_{\text{run}} - \cos k_x - \cos k_y), \quad (2)$$

with  $\alpha_{\text{PRR}} = -\alpha_{\text{run}}$ . It is used to construct the *overcomplete Wannier (OW) basis*; a spatial domain-wall Hamiltonian is not diagonalized.

For center  $\mathbf{r}$  the normalized state used by both dynamics is

$$|\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle = \frac{M_{n_{\text{shell}}}(\mathbf{r}' - \mathbf{r})M_{\text{DW}}(r'_x; r_x)|W_{\mathbf{r},\nu,\pm}\rangle}{\|M_{n_{\text{shell}}}M_{\text{DW}}|W_{\mathbf{r},\nu,\pm}\rangle\|}. \quad (3)$$

The shell mask depends on relative periodic displacement. The optional hard-wall mask allows support only in the domain containing the center. All cases use  $N_x = 20$ , walls  $x = 5, 15$ ,  $\alpha_{\text{run},\text{out}} = 30$ , trial basis  $X$ , and a maximally mixed initial state. These explicit walls override incompatible historical defaults.

Define

$$\hat{\chi}_{\mathbf{r},\nu,\pm} = \sum_{\mathbf{r}',\mu} \widetilde{W}_{\mathbf{r},\nu,\pm}(\mathbf{r}', \mu) \hat{c}_{\mathbf{r}',\mu}, \quad (4)$$

$$\hat{N}_{\mathbf{r},\nu,\pm} = \hat{\chi}_{\mathbf{r},\nu,\pm}^\dagger \hat{\chi}_{\mathbf{r},\nu,\pm}, \quad (5)$$

and the rank-one single-particle matrix  $P_{\mathbf{r},\nu,\pm} = |\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle\langle\widetilde{W}_{\mathbf{r},\nu,\pm}|$ . Distinct OW stabilizers need not commute.

## III. THE TWO REQUESTED DYNAMICS

### A. Ordered perfect-correction channel

The published correlation convention is

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{c}_i^\dagger \hat{c}_j). \quad (6)$$

The canonical class returns  $Q = 2G - 1$ ; the runner records this bridge and stores physical  $G$ . For a single mode  $P$ , let  $\mathcal{D}^\dagger[\hat{L}]$  denote the adjoint Lindblad action on  $G$ . The dephasing-free loss and gain terms are

$$\mathcal{D}^\dagger[\hat{\chi}](G) = -\frac{1}{2}\{P, G\}, \quad (7)$$

$$\mathcal{D}^\dagger[\hat{\chi}^\dagger](G) = P - \frac{1}{2}\{P, G\}. \quad (8)$$

Thus  $\mathcal{I} + \mathcal{D}^\dagger[\hat{\chi}]$  and  $\mathcal{I} + \mathcal{D}^\dagger[\hat{\chi}^\dagger]$  give the expressions sometimes called a “no-dephasing unit-step channel.” They are formal forward-Euler increments, not physical local maps for overlapping OW modes at unit step.

Number measurement contributes  $\mathcal{D}^\dagger[\hat{N}](G) = PGP - \frac{1}{2}\{P, G\}$ . The exact empty and filled reset maps are

$$\mathcal{R}_{P,+}(G) = G - \{P, G\} + PGP, \quad (9)$$

$$\mathcal{R}_{P,-}(G) = G + P - \{P, G\} + PGP, \quad (10)$$

Writing  $\mathcal{I}$  for the identity superoperator, these maps obey

$$\mathcal{R}_{P,+} - \mathcal{I} = \mathcal{D}^\dagger[\hat{\chi}] + \mathcal{D}^\dagger[\hat{N}], \quad (11)$$

$$\mathcal{R}_{P,-} - \mathcal{I} = \mathcal{D}^\dagger[\hat{\chi}^\dagger] + \mathcal{D}^\dagger[\hat{N}]. \quad (12)$$

Equivalently, each reset is exactly  $\mathcal{R}_a = \mathcal{I} + \mathcal{L}_a$ , with  $\mathcal{L}_a$  given by the appropriate right-hand side of Eq. (12). At each visited unit cell the canonical order is  $A+$  (loss),  $A-$  (gain),  $B+$  (loss),  $B-$  (gain); cells are visited in the saved random order. The full sweep is the corresponding noncommuting product

$$\mathcal{E}_s = \prod_{\mathbf{r} \in s}^{\leftarrow} \mathcal{R}_{P_{\mathbf{r}}, B, -, -} \mathcal{R}_{P_{\mathbf{r}}, B, +, +} \mathcal{R}_{P_{\mathbf{r}}, A, -, -} \mathcal{R}_{P_{\mathbf{r}}, A, +, +}, \quad (13)$$

with schedule record  $s$ . Equation (13) is the production channel. It is also exact to write

$$\mathcal{E}_s = \mathcal{I} + \mathcal{L}_{\text{cycle}, s}, \quad \mathcal{L}_{\text{cycle}, s} = \mathcal{E}_s - \mathcal{I}. \quad (14)$$

Here  $\mathcal{L}_{\text{cycle}, s}$  contains every schedule-ordered cross term generated by the product; it is not  $\sum_a \mathcal{L}_a$ . The literal no-dephasing unit-step Euler product is excluded because it is not positivity preserving for overlapping OW modes; it is retained only as a labelled small-system diagnostic.

### B. Infinitesimal continuous equation

Let  $a = (\mathbf{r}, \nu, n)$ ,  $P_a = P_{\mathbf{r}, \nu, n}$ ,  $s_a = 1$  for  $n = -$ , and  $s_a = 0$  for  $n = +$ . The continuous perfect-correction action matched to the local increments above is

$$\begin{aligned} \frac{dG}{dt} &= \mathcal{L}_{\text{PC}}^{(\zeta)}(G) \\ &= \sum_a \left[ s_a P_a - \frac{1}{2}\{P_a, G\} + \zeta P_a G P_a - \frac{\zeta}{2}\{P_a, G\} \right]. \end{aligned} \quad (15)$$

The  $\zeta = 0$  arm contains unit gain and loss. The  $\zeta = 1$  arm additionally contains unit number dephasing, equivalent at the many-body level to adding  $\sum_{\mathbf{r}, \nu, n} \mathcal{D}[\hat{N}_{\mathbf{r}, \nu, n}]$ . Although number dephasing need not preserve Gaussian many-body states, Eq. (15) closes exactly on  $G$ . Consequently an entropy evaluated only from the natural occupations is called a Gaussian entropy proxy whenever  $\zeta = 1$ . Equation (15) is the infinitesimal Poisson generator  $\sum_a \mathcal{L}_a$  (or its dephasing-deleted version), not the finite-sweep  $\mathcal{L}_{\text{cycle}, s}$  of Eq. (14).

The production integrator is complex128 RK4 with  $\delta t = 0.05$ . A separate representative validation subset at  $\delta t = 0.1, 0.05, 0.025$  forms the step-convergence gate; these are not additional production physics cases. The file `CI.Lindblad_DW.py` is an algebraic reference only; production consumes the same OW vectors as the channel and does not inherit that file’s distinct wall rule.

## IV. ORDER, TRANSLATION, AND MOMENTUM ESTIMATORS

Because Eq. (3) is translated identically at every  $y$  and the wall mask depends only on  $x$ ,

$$[T_y, \mathcal{L}_{\text{PC}}^{(\zeta)}] = 0. \quad (16)$$

A translation-invariant initial state remains invariant under the continuous equation, and  $G(k_y)$  has exact  $2N_x \times 2N_x$  blocks. A finite ordered product generally does not commute with  $T_y$ , even if every local family is covariant. We therefore do not diagonalize the diagonal Fourier slices of the untwirled channel state and call them bands.

For each independent schedule sample  $s$ , first form the approved late-cycle average

$$\overline{G}_s = \frac{1}{N_y} \sum_{c=N_y+1}^{2N_y} G_s(c), \quad (17)$$

then the exact spatial sliding estimator

$$\mathcal{T}_y[\overline{G}_s] = \frac{1}{N_y} \sum_{q=0}^{N_y-1} T_y^q \overline{G}_s T_y^{-q}. \quad (18)$$

Its blocks define the channel’s explicitly *twirled* spectrum. Every such plot is accompanied by the untwirled residual

$$\varepsilon_T = \frac{\|T_y \overline{G}_s T_y^{-1} - \overline{G}_s\|_F}{\|\overline{G}_s\|_F}. \quad (19)$$

For a complementary representation that makes no symmetry assumption, a global mode  $G|\phi_j\rangle = \nu_j|\phi_j\rangle$  carries momentum weight

$$Z_j(k_y) = \sum_{x, \mu} |\langle x, k_y, \mu | \phi_j \rangle|^2, \quad \sum_{k_y} Z_j(k_y) = 1. \quad (20)$$

## V. CAMPAIGN AND STORAGE DESIGN

Cheap observables are retained for sample  $s$  at every integer cycle  $0, 1, \dots, 2N_y$ . Spectral estimators are evaluated only at the explicit checkpoint cycles declared in the immutable configuration; a dense  $2560 \times 2560$  matrix is not diagonalized after every cycle. Dense states are checkpoint-resolved, with final and late-average matrices retaining the sample axis. Each atomic case directory contains an NPZ product, JSON metadata, a terminal marker, and checksums. No pre-existing run directory is overwritten.

TABLE I. Locked production design. Alpha scans use DW truncation on only; endpoint size scans use it on and off. The independent unit is the schedule sample for the ordered channel and the deterministic run for the continuous arms.

|                     |                                                                                                                                                                                                                                                                                          |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Endpoint grid       | $N_y = 32, 48, 64$ ; $\alpha_{\text{run}, \text{in}} = 1$ ; $n_{\text{shell}} = 1, 2$ ; truncation on/off; channel with dephasing; continuous evolution without/with dephasing.                                                                                                          |
| Alpha grid          | $N_y = 64$ ; $\alpha_{\text{run}, \text{in}} = 1, 1.5, 1.75, 1.875, 2.125, 2.25, 2.5, 3$ ; $n_{\text{shell}} = 1, 2$ ; truncation on only.                                                                                                                                               |
| Full-frame controls | Continuous dynamics only; endpoint truncation on/off and static alpha scan with truncation on.                                                                                                                                                                                           |
| Timing              | Main evolution through $2N_y$ ; late average over $N_y + 1, \dots, 2N_y$ ; response through $N_y/2$ .                                                                                                                                                                                    |
| Response subset     | Exactly 22 endpoint cases: finite $n_{\text{shell}} = 1$ at all sizes, both truncation settings, and all three arms; plus full-frame continuous cases at $N_y = 64$ for both truncation settings. Alpha scans, $n_{\text{shell}} = 2$ , and the $S = 10$ schedule control omit response. |
| Random schedules    | Root seed 20260814; one matched seed in the main grid; a separate canonical $N_y = 64, n_{\text{shell}} = 1$ , truncated channel case with $S = 10$ independent schedule seeds.                                                                                                          |
| Case count          | 105 unique case specifications: 27 channel, 39 continuous without dephasing, and 39 continuous with dephasing.                                                                                                                                                                           |

## VI. OCCUPATION-SPECTRUM RESULTS AND ESTIMATORS

Let

$$G(k_y)|\phi_j(k_y)\rangle = \nu_j(k_y)|\phi_j(k_y)\rangle \quad (21)$$

denote either an exact continuous block or an explicitly twirled channel block. With  $\mathcal{W}_a = \{x_a - 1, x_a, x_a + 1\}$ ,

$$W_j^{(a)}(k_y) = \sum_{x \in \mathcal{W}_a, \mu} |\phi_{j, x\mu}(k_y)|^2. \quad (22)$$

Eligible wall modes have  $W_j^{(a)} \geq 0.25$  and minimize  $|\nu_j - 1/2| + 0.02(1 - W_j^{(a)})$  independently at each momentum. The displayed slope is an ordinary least-squares fit over  $|k_y| \leq \max(2.5 \cdot 2\pi/N_y, 0.22)$ . It is not overlap tracking or a topological index.

The global midpoint gap, bulk-excluded gap, Gaussian entropy proxy, Gaussian charge-variance proxy, and midgap wall localization are

$$\Delta_{1/2} = \min_j |\nu_j - \tfrac{1}{2}|, \quad (23)$$

$$\Delta_{\text{bulk}} = \min_{j: W_j^{\text{W}} < 0.25} |\nu_j - \tfrac{1}{2}|, \quad (24)$$

$$S_G = - \sum_j [\nu_j \ln \nu_j + (1 - \nu_j) \ln(1 - \nu_j)], \quad (25)$$

$$V_G = \text{tr}[G(\mathbf{1} - G)] = \sum_j \nu_j(1 - \nu_j), \quad (26)$$

$$\mathcal{L}_{\text{mid}} = \sum_{x \in \mathcal{W}} p_x, \quad (27)$$

where  $p_x$  is averaged over the  $2N_y$  modes closest to half occupation and normalized to unit total weight. For a non-Gaussian full Fock-space state,  $V_G$  is not the true charge variance; it is explicitly reported only as a two-point Gaussian proxy. No eigenvalue outside  $[0, 1]$  is clipped to manufacture an entropy; its physicality violation is reported instead.

Figure 1 shows the central state-level result: a pair of branches crosses  $\nu = 1/2$  near  $k_y = 0$ ,

while the bulk-excluded midpoint gap remains 0.496–0.499. For  $n_{\text{shell}} = 1$  with DW truncation, the selected wall slopes are  $(-0.5858, +0.5843)$  for the channel,  $(-0.6599, +0.6599)$  for continuous evolution without dephasing, and  $(-0.6186, +0.6186)$  with dephasing. The channel's finite-size midpoint gap is  $1.74 \times 10^{-2}$ ; the continuous values are  $3.33 \times 10^{-10}$  and  $3.12 \times 10^{-10}$ . The continuous full-frame slopes are  $\pm 0.9496$  without and  $\pm 0.9003$  with dephasing. Finite shell changes the slope and localization, but not the existence of the crossing.

The occupation gap grows sharply through the unsampled  $\alpha = 2$  reference. For the  $n_{\text{shell}} = 1$  dephasing-on continuous arm it changes from  $7.04 \times 10^{-4}$  at  $\alpha = 1.5$  to  $5.75 \times 10^{-2}$  at 1.75,  $1.68 \times 10^{-1}$  at 1.875, and  $3.89 \times 10^{-1}$  at 2.125; the channel shows the same crossover with larger finite-sweep irregularity. In contrast, the measured relaxation scales remain finite: approximately 2.62 or larger for the channel, 1.00 or larger without continuous dephasing, and 1.90 or larger with it. The campaign therefore finds an occupation-spectrum transition but no dynamical gap closing in the measured sector.

## VII. RELAXATION AND DENSITY-KICK RESPONSE

The affine continuous equation has a homogeneous covariance generator  $\mathbb{L}^{(\zeta)}$ ; the ordered sweep has a homogeneous map  $\mathbb{M}_{\mathbf{s}}$ . Their comparable relaxation scales are

$$\gamma_{\text{cont}} = \min_{\ell \neq 0} [-\text{Re } \ell], \quad (28)$$

$$\gamma_{\text{ch}} = \min_{0 < |m| < 1} [-\ln |m|], \quad (29)$$

with exact neutral sectors removed by a stated tolerance. Complex and nonnormal spectra are retained rather than replaced by the older  $2d_{\text{min}}$  formula.

A source at wall  $a$  and coordinate  $y_0$  is kicked by

$$\hat{U}_\epsilon = \exp[-i\epsilon \hat{n}_{a, y_0}], \quad \delta G(0) = i[P_{a, y_0}, G_{\text{bg}}]. \quad (30)$$

The same homogeneous channel or generator then propagates the tangent. The wall profile and positive-lobe

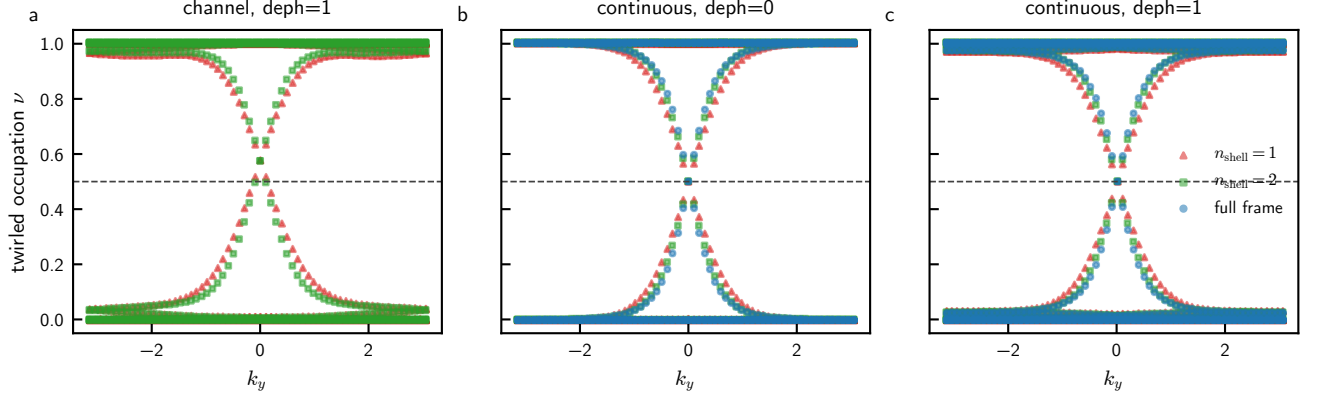


FIG. 1. Canonical  $k_y$ -resolved natural occupations at  $N_x \times N_y = 20 \times 64$ ,  $\alpha_{\text{run},\text{in}} = 1$ ,  $\alpha_{\text{run},\text{out}} = 30$ , maximal-mixing initialization, and DW-support truncation on. Red triangles and green squares are  $n_{\text{shell}} = 1, 2$ ; blue circles are the continuous full-frame control. The channel panel is the exact translation twirl of the cycle-65–128 average for one matched random schedule; continuous panels use exact  $k_y$  blocks over the corresponding time window. The dashed line marks  $\nu = 1/2$ .

center are

$$\chi_a^{\text{wall}}(r, t) = \sum_{x \in \mathcal{W}_a, \mu} [\delta G(t)]_{x, y_0+r, \mu; x, y_0+r, \mu}, \quad (31)$$

$$y_{+,a}(t) = \frac{\sum_r \tilde{r}[\chi_a^{\text{wall}}(r, t)]_+}{\sum_r [\chi_a^{\text{wall}}(r, t)]_+}, \quad (32)$$

where  $\tilde{r} = ((r + N_y/2) \bmod N_y) - N_y/2$ . Velocity is the OLS slope over  $2 \leq t \leq 0.375N_y$ , subject to the saved response-norm threshold. For the ordered channel, all  $y_0$  are retained before averaging, so the schedule-origin spread is a deterministic systematic rather than a statistical confidence interval.

The continuous response is counterpropagating: velocities are  $(-0.0989, +0.0989)$  without dephasing and  $(-0.1557, +0.1557)$  with dephasing, with minimum deterministic  $R^2 = 0.9983$  and  $0.9953$ , respectively. The ordered channel does not pass the analogous directional test. Its mean source-resolved velocities are  $(-0.5999, -0.6643)$ , both with the same sign, and the minimum  $R^2$  values are only  $5 \times 10^{-6}$  and  $2.7 \times 10^{-4}$ . The channel perturbation also decays before most of the nominal fit window. We therefore do not interpret its fitted displacement as a chiral velocity.

The size and direct-dynamics checks in Fig. 4 replace the old auxiliary finite- $p$  Strang comparison. The physical channel has one unit reset per stabilizer, so no weak-step order is inferred from it.

The finite ordered sweep does not approach the infinitesimal generator over the three available sizes. Across the finite-shell endpoint controls, the late-state relative Frobenius difference is 2.29–3.02% and the occupation

RMS difference is 0.0072–0.0091; the response-profile relative error is 1.06–1.28 wherever it is evaluated. This is expected because a unit sweep retains schedule-ordered cross terms that are absent from the summed infinitesimal generator.

## VIII. VALIDATION AND INTERPRETATION RULES

All 105 cases completed and all immutable result hashes survived analysis. The local equations agree with the canonical channel on small systems; the continuous action agrees with the derivative of weakened local maps; RK4 shows the expected step convergence; the full-dephasing channel remains physical; continuous translation residuals meet tolerance; cycle coordinates and sample IDs are complete; and the response tangent agrees with a finite central difference. Three stored gauge-dependent raw-OW byte hashes differed across otherwise matched arms; fresh construction reproduced one recorded realization in each group, so these are retained as provenance warnings rather than misidentified as physical-projector mismatches.

The completed note may report stationary or late-state occupation flow, relaxation, and damped directional response. It may not infer a trajectory Lyapunov boundary mode, quantized transport, or schedule-independent chirality. The random ordered channel’s translation twirl is an estimator applied to its late state, not a symmetry retroactively assigned to the underlying map.

This source is maintained under a strict nine-page main-text ceiling. Detailed schemas, derivations, convergence tables, and legacy reconciliation belong in the companion supplement and analysis notebook.

[1] A. Bhuiyan, G. Pan, and S.-K. Jian, “Free-fermion dynamics with measurements: Topological classification and

adaptive preparation of topological states,” Phys. Rev.

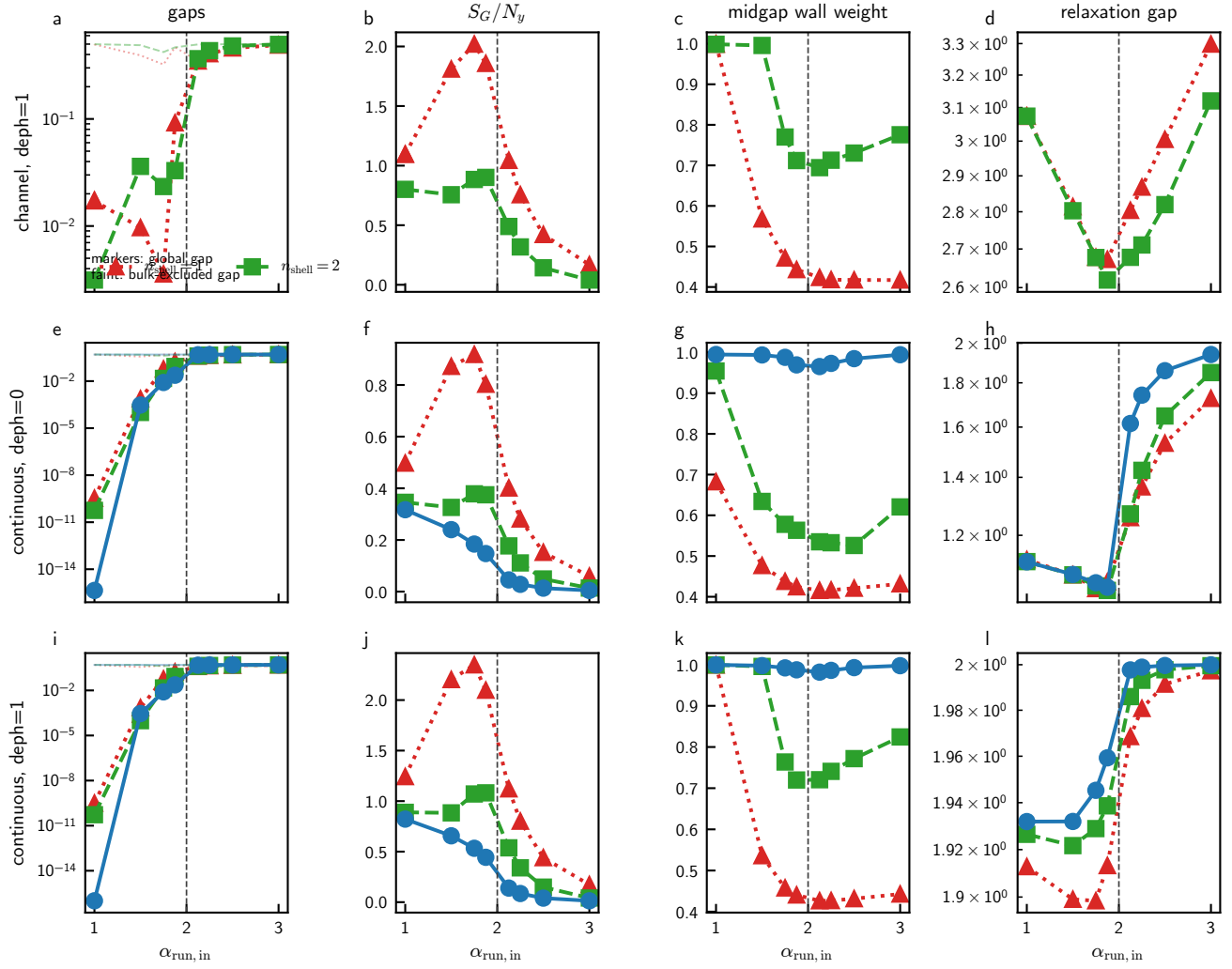


FIG. 2. Matched alpha scan at  $N_y = 64$  with DW-support truncation on. Columns report the global (markers) and bulk-excluded (faint curves) midpoint gaps,  $S_G/N_y$ , midgap wall localization, and relaxation. The vertical line at  $\alpha_{\text{run},\text{in}} = 2$  is a parent-model reference, not a sampled point. Rows are the full-dephasing channel, continuous no-dephasing evolution, and continuous full-dephasing evolution; colors/markers distinguish shell range.

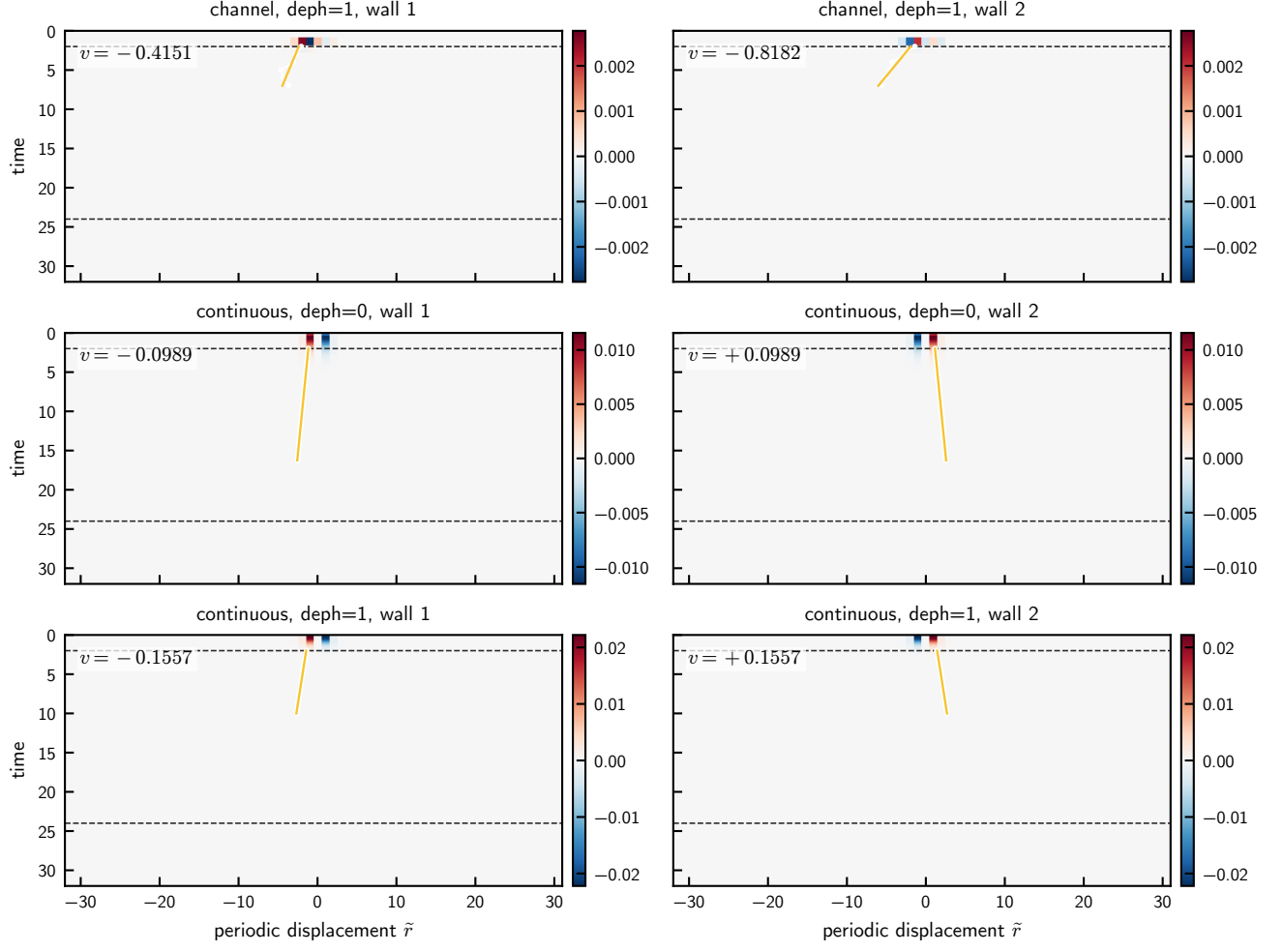


FIG. 3. Local density-phase response for the canonical  $N_y = 64$ ,  $n_{\text{shell}} = 1$ , DW-truncated endpoint. Rows compare the channel and the two continuous arms; columns are the two walls. The channel stores all 64 source origins and is source-averaged only for display; exact translation symmetry makes one source sufficient in the continuous arms. White points and gold lines show the retained positive-lobe centers and OLS fits; dashed lines mark the nominal fit window. Displayed annotations are profile slopes, while source-resolved channel fit quality is reported in the text.

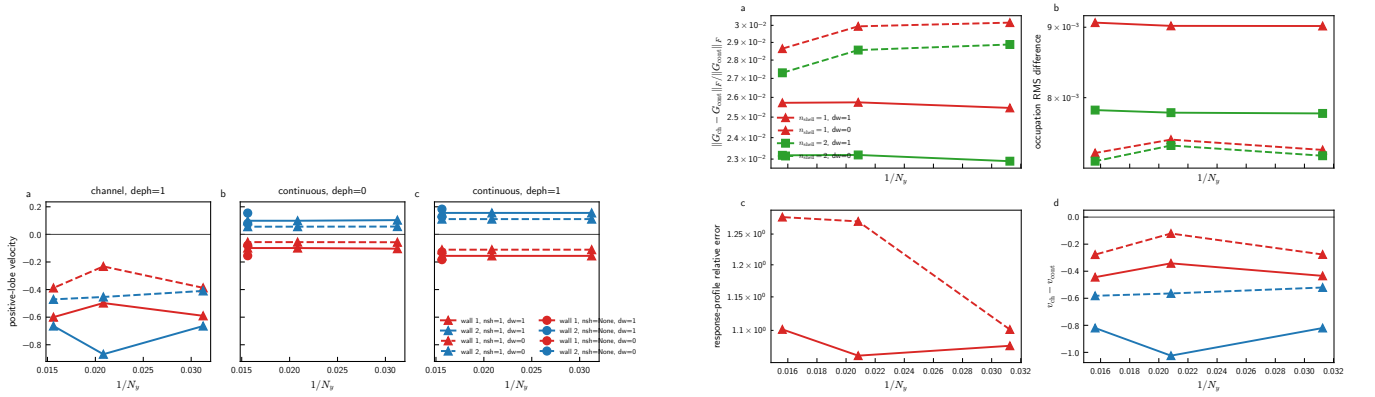


FIG. 4. Endpoint checks. Left: raw fitted velocities versus  $1/N_y$  for both walls and both DW-support settings. Right: finite unit-step channel minus the dephasing-matched continuous result, including state, occupation, response-profile, and velocity differences. No thermodynamic extrapolation or weak-step convergence fit is performed.